
3. 5G Mobile Communication System Developer Guide



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Chapter 1: Introduction

1.1 Purpose and Intended Audience

1.1.1 Purpose

This guide aims to provide comprehensive direction for developers building cutting-edge, high-performance, and feature-rich 5G network systems. It is especially intended to support the following areas:

- UE (User Equipment) design and protocol stack implementation
- gNB (Next Generation Node B) construction and performance optimization
- 5GC (5G Core) service design and network slicing

By leveraging this guide, readers will acquire the knowledge and tools necessary for the efficient development and operation of a complete 5G system.

1.1.2 Intended Audience

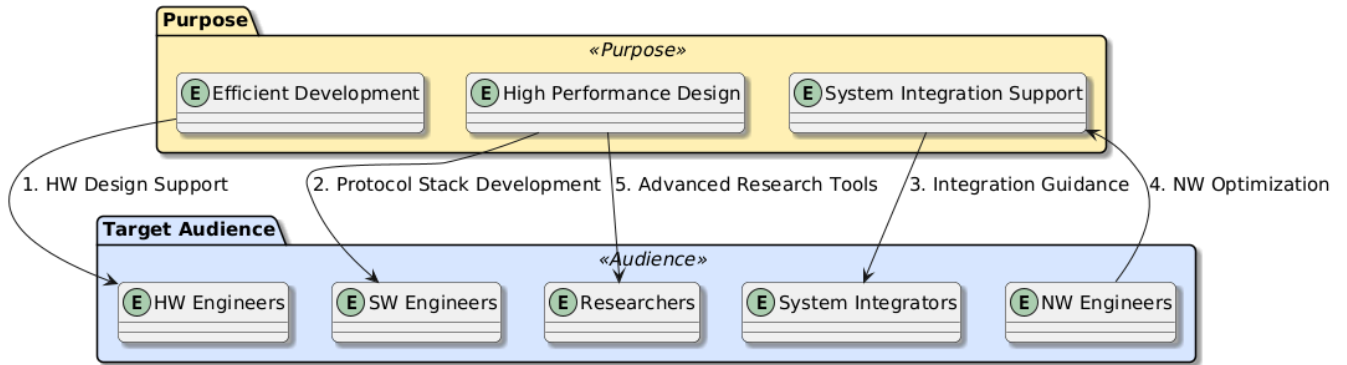
This guide is targeted at engineers and technical professionals involved in 5G network system development, including:

- **HW Engineers:**
Responsible for processor, memory, and antenna design
- **SW Engineers:**
Responsible for implementing the 5G protocol stack and related drivers
- **System Integrators:**
Handle integration and testing of UE, gNB, and 5GC
- **Network Engineers:**
Focus on network slicing and operational optimization
- **Researchers:**
Explore next-generation 5G technologies

This guide provides practical information and examples tailored to the specific roles and responsibilities of each of these audiences.

Below is a diagram illustrating the relationship between the guide's purpose and its intended audience:

1.1 Purpose and Intended Audience



1.1.3 Explanation

- The Purpose package divides the guide's objectives into three key categories: Efficient Development, High Performance Design, and System Integration Support.
- The Target Audience package lists the key professionals involved in 5G system development, grouped by expertise.
- Relationships between the objectives and audiences are clearly visualized with arrows and numbered annotations.
- The color-coded grouping makes it easier to understand at a glance.

By using this diagram, readers can quickly identify the scope and applicability of the guide in relation to their own roles.

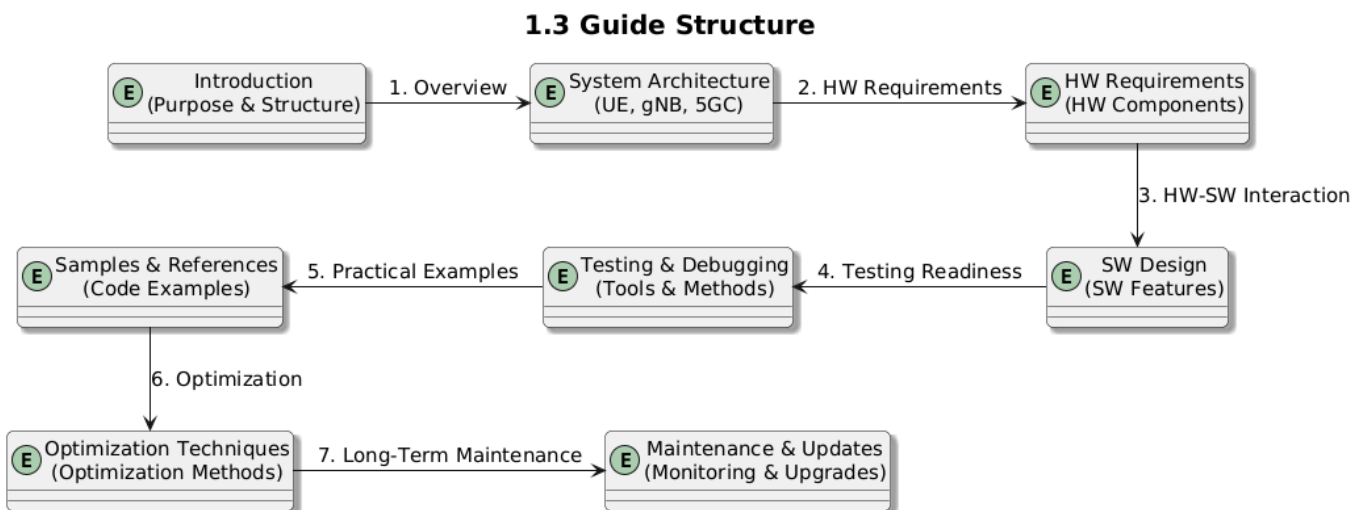
1.2 Guide Structure

This guide is organized into the following sections. Each section corresponds to a specific stage in efficiently designing, implementing, testing, and optimizing a 5G network (NW) system:

- **Introduction**
Explains the guide's purpose, intended audience, an overview of 5G NW systems, and the overall structure of the guide.
- **System Architecture**
Describes the core components of the 5G network (UE, gNB, 5GC) and their interactions.
- **Hardware Requirements**
Details the hardware requirements needed for designing each component, including processors, memory, antennas, and more.
- **Software Design**
Covers 5G protocol stack implementation, network slicing, driver design, and other software-related topics.
- **Testing & Debugging Methods**
Provides a comprehensive explanation of test tools and debugging techniques that developers can use.
- **Sample Code & Reference Implementations**
Offers practical implementation examples for each component.
- **Optimization Techniques**
Introduces methods for achieving low-latency and high-efficiency system designs.
- **Maintenance & Updates**
Discusses maintainability and update strategies for long-term operation.

Diagram: Guide Structure

The diagram below visually represents the structure of this guide:



Explanation

- Each section is displayed as a separate entity, representing a different colored package (visualized through grouping in diagrams).
- Arrows with numbered steps clearly define the relationships and sequence of stages.
- This structure helps developers intuitively understand the flow of the development process from start to finish.

By referencing this structure, developers can gain a comprehensive and efficient understanding of the guide's contents and the logical progression of 5G system development.

Chapter 2: Overall System Architecture

2.1 Overview of 5G Network Components

UE (User Equipment)

The UE is the end-user device that accesses the 5G network. It includes the following key functionalities:

- **Communication Module:**
Handles communication protocols from the PHY layer up to the application layer.
- **RF Design:**
Optimized antenna systems for efficient wireless transmission and reception.
- **Power Optimization:**
Maximizes battery life for mobile devices through power-efficient design.

gNB (Next-generation Node B)

The gNB is the core element of the 5G Radio Access Network (RAN), responsible for:

- **PHY Layer:**
Manages signal encoding, modulation, and scheduling.
- **Data Processing:**
Provides efficient processing mechanisms to support high-speed data transfer.
- **User and Control Plane Separation:**
Enhances network flexibility by separating the U-plane and C-plane.

5GC (5G Core Network)

The 5GC is the central component that controls and delivers services across the 5G network. It includes:

- **Network Slicing:**
Efficiently manages multiple virtual networks on a shared infrastructure.
- **Mobility Management:**
Supports user mobility and seamless roaming.
- **QoS Control:**
Ensures high-speed, low-latency, and reliable communication.

Diagram: Overview of 5G Network Components

The following diagram visualizes the three core components and their interactions:

2.1 Overview of 5G Network Components

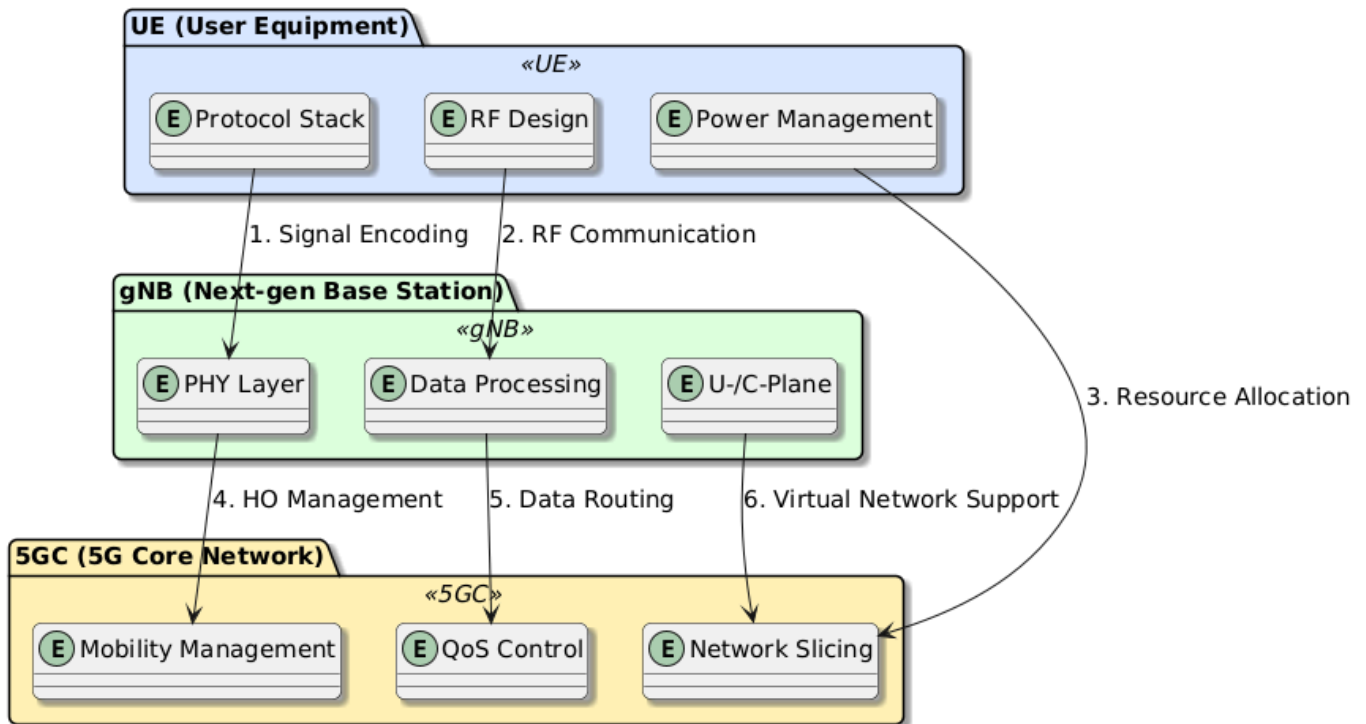


Diagram Explanation

- Each component (UE, gNB, 5GC) is color-coded for visual distinction.
- Arrows between entities are numbered to show sequential interactions.
- The flow of processing is clearly represented by directional arrows, emphasizing the cooperation among components.

This diagram illustrates how the core components of the 5G network interact to enable efficient communication. It helps developers deepen their understanding of the system as a whole.

2.2 Interfaces Between Architectures

In a 5G network system, the key interfaces between core architectural components are as follows:

UE ↔ gNB (Uu Interface)

- Provides the wireless communication link between UE and gNB.
- Handles data transmission and reception via PHY, MAC, and RLC layers.
- Transfers control messages for authentication, resource allocation, and handover (HO).

gNB ↔ gNB (Xn Interface)

- Manages inter-base-station handover and resource coordination.
- Supports the exchange of both User Plane (U-Plane) and Control Plane (C-Plane) data.

gNB ↔ 5GC (NG Interface)

- Manages both user data traffic and control signaling.
- Separated into NG-C (Control Plane) and NG-U (User Plane).
- Handles session management, QoS control, and user authentication.

Diagram: Architecture Interfaces

The diagram below visualizes these interfaces and their respective roles:

2.2 Architecture Interfaces

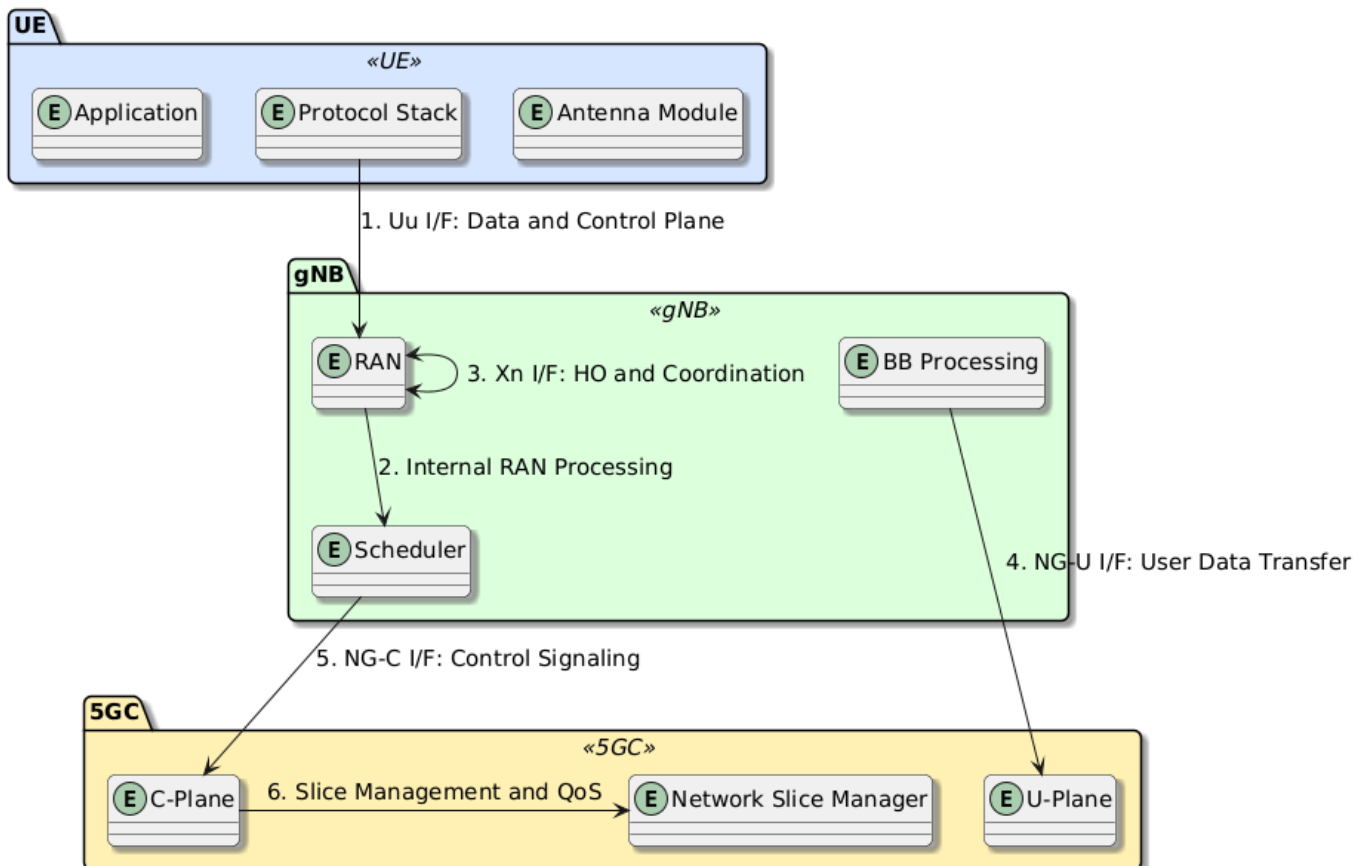


Diagram Explanation

UE

Includes the application layer, protocol stack, and antenna module. Responsible for communication with the gNB.

gNB

Manages the Radio Access Network (RAN), including scheduling and baseband (BB) processing.

5GC

Controls and manages the entire network via the Control Plane and User Plane.

Interface Arrows

Each arrow is labeled with the direction of data or control flow and its function. Numbers indicate the sequence of processing or interaction.

This architecture design helps developers clearly understand the functional roles and responsibilities of each interface between the UE, gNB, and 5GC.

2.3 Requirements Based on 3GPP Specifications

The 3GPP specifications provide essential standards for the design and implementation of 5G network systems. This section outlines the key requirements defined by 3GPP across the main components of the system.

1. Requirements for the UE (User Equipment)

- **Conformance to 5G Protocol Stack:**
Must support functionalities for 5GMM, 5GSM, RRC, SDAP, PDCP, RLC, MAC, and PHY layers.
- **RF Performance:**
Must comply with frequency bands and spectral efficiency criteria defined in 3GPP Releases 15 and 16.
- **Mobility Management:**
Should meet the criteria for handover (HO) and cell reselection procedures.

2. Requirements for gNB (Next-Generation Node B)

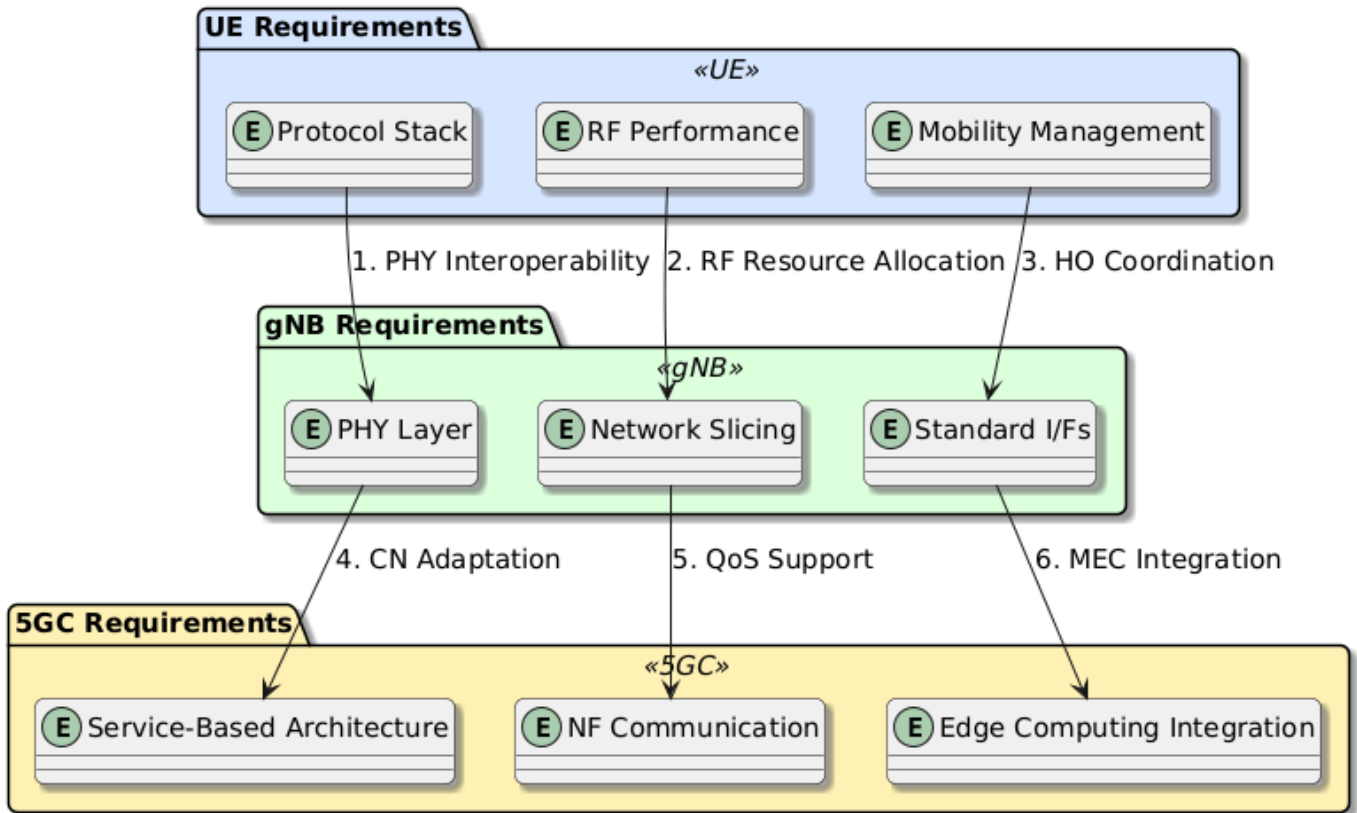
- **PHY Layer Processing:**
Must support advanced features such as Massive MIMO and Beamforming.
- **Network Slicing:**
Should enable flexible resource allocation based on Quality of Service (QoS) requirements.
- **Standard Interfaces:**
Must ensure compatibility with NG interfaces (between gNB and 5GC) and Xn interfaces (between gNBs).

3. Requirements for the 5GC (5G Core Network)

- **Service-Based Architecture (SBA):**
Must follow a modular Core Network design in compliance with 3GPP standards.
- **NF Communication:**
Must enable standardized communication between Network Functions such as AMF, SMF, and UPF.
- **Edge Computing:**
Should support integration with Multi-access Edge Computing (MEC) systems.

Diagram: 3GPP Requirements

The following diagram illustrates how these 3GPP requirements apply across the 5G system components.

2.3 3GPP Requirements**Diagram Explanation****UE**

The UE's key requirements—protocol stack conformance, RF performance, and mobility management—are represented as entities. Their interoperability with the gNB is shown through direct connections.

gNB

The gNB's core requirements include PHY layer support, network slicing capability, and standard interface compliance. Each of these is connected to relevant UE and 5GC requirements to show end-to-end functionality.

5GC

Key 5GC requirements such as SBA, NF communication, and edge computing integration are illustrated as entities, emphasizing modularity and service orientation.

By using this diagram, developers can visually understand how 3GPP specifications are applied across the entire 5G system architecture, ensuring standard-compliant design and implementation.

Chapter 3: Hardware Requirements

3.1 Processor Selection Guidelines

In 5G network systems, selecting the appropriate processor for each key component—UE, gNB, and 5GC—is critical for ensuring optimal performance, efficiency, and scalability. This section outlines the key guidelines for processor selection.

1. Performance Requirements

- **Computing Power:**

Multi-core architectures such as ARM Cortex-A or RISC-V are essential for achieving high throughput.

- **Low-Latency Processing:**

Especially for gNB, real-time responsiveness is crucial and must be ensured through low-latency processor design.

2. Power Consumption

- **Power Efficiency:**

For UE devices, low-power processors are critical to extend battery life.

- **Dynamic Power Management:**

Support for Dynamic Voltage and Frequency Scaling (DVFS) is required to balance performance with power efficiency.

3. Scalability and Flexibility

- **Use-Case Optimization:**

Different components demand different capabilities—e.g., high parallel processing in 5GC, fast data processing in gNB.

- **Hardware Acceleration:**

Support for AI inference and packet processing acceleration is beneficial for performance optimization.

4. Compatibility and Future-Proofing

- **Software Compatibility:**

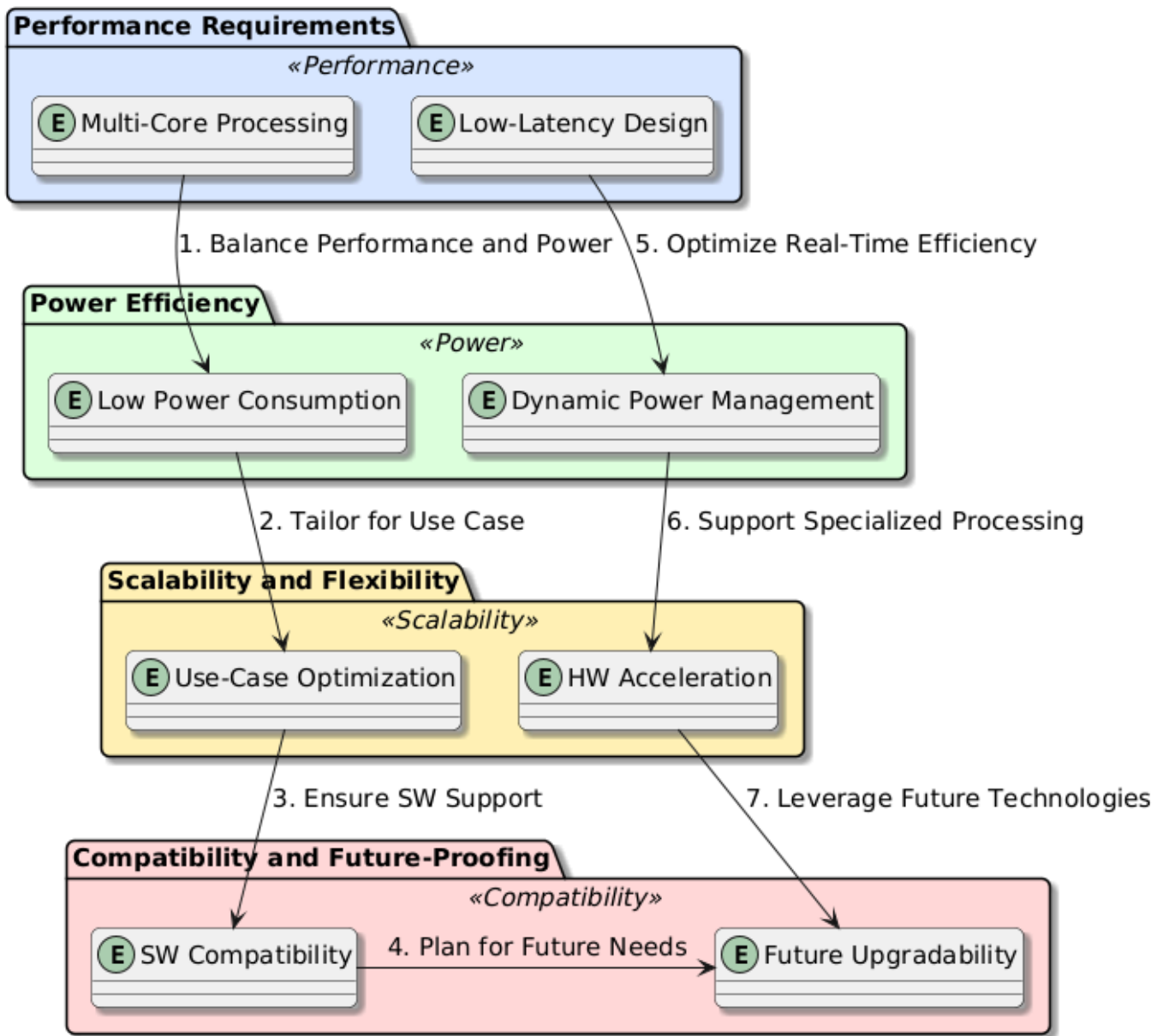
Ensuring compatibility with Linux kernels and RTOS platforms is vital.

- **Future Upgradability:**

The processor must allow for potential upgrades in preparation for 5G Advanced and future 6G technologies.

Diagram: Processor Selection Guidelines

The following diagram visually represents the multidimensional aspects of processor selection.

3.1 Processor Selection Guidelines**Diagram Explanation****Performance Requirements**

The diagram highlights “Multi-Core Processing” and “Low-Latency Design” as core performance entities, emphasizing the importance of computing efficiency.

Power Consumption

“Low Power Consumption” and “Dynamic Power Management” are illustrated to stress the importance of power optimization, particularly in mobile devices.

Scalability and Flexibility

By incorporating “Use-Case Optimization” and “Hardware Acceleration,” the diagram reflects the need for flexible designs that accommodate diverse workloads across components.

Compatibility and Future-Proofing

“Software Compatibility” and “Future Upgradability” ensure that the processor design supports current platforms and is ready for future 5G evolution.

By using this diagram, developers can comprehensively assess processor requirements from multiple perspectives and make well-informed design choices for 5G systems.